

Year 7 • Chapter 5A • Algebra: answers (teacher copy)

5.01 The commutative and associative laws (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 28
- Ex 1b: 118
- Ex 2a: 170
- Ex 2b: 3 100
- Ex 3a: 390
- Ex 3b: 2 800

Easy (drill)

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|----------|-----------|-----------|
| 1. 260 | 12. 41000 | 23. false |
| 2. 4100 | 13. 220 | 24. false |
| 3. 680 | 14. 4000 | 25. false |
| 4. 300 | 15. 310 | 26. true |
| 5. 30 | 16. 3400 | 27. true |
| 6. 2800 | 17. false | 28. true |
| 7. 1600 | 18. false | 29. false |
| 8. 80 | 19. false | 30. true |
| 9. 2600 | 20. false | 31. false |
| 10. 2900 | 21. false | 32. false |
| 11. 300 | 22. false | |

Medium (drill)

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|--------|---------|-----------------|
| 1. 132 | 8. 104 | 15. 103 |
| 2. 85 | 9. 119 | 16. 126 |
| 3. 114 | 10. 101 | 17. associative |
| 4. 60 | 11. 124 | 18. commutative |
| 5. 134 | 12. 101 | 19. associative |
| 6. 158 | 13. 49 | 20. commutative |
| 7. 95 | 14. 117 | |

Challenging

- No: $10 - 4 = 6$ but $4 - 10 = -6$
- No: $(24 \div 6) \div 2 = 2$ but $24 \div (6 \div 2) = 8$
- 17 000

Extension

- 90 000
- $38 + 62 = 100$ and $47 + 53 = 100$

Exit ticket

- Level 1: 47
- Level 2: 230
- Level 3: 1 300

5.02 The distributive law (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: 198
- Ex 1b: 136
- Ex 2a: 297
- Ex 2b: 156
- Ex 3a: 288
- Ex 3b: 3 465

Easy (drill)

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|--------|---------|----------|
| 1. 63 | 8. 147 | 15. 140 |
| 2. 475 | 9. 378 | 16. 648 |
| 3. 111 | 10. 495 | 17. 4257 |
| 4. 320 | 11. 828 | 18. 4455 |
| 5. 272 | 12. 756 | 19. 1450 |
| 6. 144 | 13. 135 | 20. 741 |
| 7. 582 | 14. 333 | 21. 594 |

22. 273
23. 117
24. 266
25. 2376

26. 1980
27. 540
28. 418
29. 4851

30. 1479
31. 3465
32. 777

Medium (drill)

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|-------|---------------|---------------|
| 1. 50 | 7. 50 | 13. $4k - 8$ |
| 2. 50 | 8. 80 | 14. $8k - 88$ |
| 3. 2 | 9. $4n - 36$ | 15. $2c - 12$ |
| 4. 1 | 10. $4t - 16$ | 16. $2t + 12$ |
| 5. 2 | 11. $5c - 35$ | |
| 6. 60 | 12. $4y - 12$ | |

Challenging

- $7 \times 100 = 700$
- $15 \times 10 = 150$
- e.g. $18 \times 25 = 10 \times 25 + 8 \times 25 = 450$; or $20 \times 25 - 2 \times 25 = 450$

Extension

- 64 064
- 3 700

Exit ticket

- Level 1: 84
- Level 2: 114
- Level 3: 4 590

5.03 Variables and algebraic notation

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: $a - (-a) = 2 \times a$
- Ex 1b: $4x$
- Ex 2a: $10a$
- Ex 2b: $24abc$
- Ex 3a: $4 \times m \times n \times k$
- Ex 3b: $9 \times c \times c$; $6 \times a \div b$

Easy (drill)

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|-----------|----------------------------|---------------------------|
| 1. $8b$ | 12. $9a$ | 23. $9 \times k \times p$ |
| 2. $11a$ | 13. p^2 | 24. $11 \times k$ |
| 3. $7py$ | 14. $6a$ | 25. $3 \times m$ |
| 4. $4p$ | 15. k^2 | 26. $8 \times k \times n$ |
| 5. $6t$ | 16. $3m$ | 27. $3 \times c$ |
| 6. a^2 | 17. $10 \times t \times x$ | 28. $6 \times m$ |
| 7. m^2 | 18. $12 \times c$ | 29. $5 \times c \times t$ |
| 8. $12ky$ | 19. $6 \times m \times x$ | 30. $4 \times p$ |
| 9. $4k$ | 20. $7 \times c$ | 31. $2 \times a \times p$ |
| 10. $11a$ | 21. $11 \times m$ | 32. $9 \times a \times b$ |
| 11. $6c$ | 22. $3 \times a$ | |

Medium (drill)

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|------------------|-------------------|---|
| 1. $4b^3n$ | 8. bm^2 | 15. $\frac{c}{7}$ |
| 2. $4k^3n$ | 9. $6a^3x$ | 16. $\frac{c}{8}$ |
| 3. $9n^2y$ | 10. kn^2 | 17. $4 \times b \times b \times p$ |
| 4. $\frac{m}{7}$ | 11. ax^2 | 18. $6 \times p \times p \times t$ |
| 5. mp^2 | 12. $\frac{c}{5}$ | 19. $4 \times a \times a \times x$ |
| 6. $\frac{c}{4}$ | 13. $5n^3p$ | 20. $3 \times b \times b \times m$ |
| 7. bn^2 | 14. $5a^3y$ | 21. $5 \times k \times k \times k \times p$ |
| | | 22. $3 \times k \times k \times m$ |

Challenging

- $a + a = 2 \times a$
- $a \times 0 = 0$

3. $3n + 1$ sticks

Extension

1. $n \times n - (n - 1)(n + 1) = 1$
2. check

Exit ticket

1. Level 1: $3y$
2. Level 2: $12pq$
3. Level 3: $5 \times a \times a \times b$

5.04 Expressions from words (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: $3N$
2. Ex 1b: $N - 5$
3. Ex 2a: $N + 1$
4. Ex 2b: $\frac{N}{3}$
5. Ex 3a: $2H - k$
6. Ex 3b: $2(x + 4)$

Easy (drill)

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|-------------|-------------------|--------------------|
| 1. $b - 2$ | 10. $\frac{a}{2}$ | 18. $n + 6$ |
| 2. $2c$ | 11. $\frac{b}{7}$ | 19. $n - 6$ |
| 3. $c - 3$ | 12. $a - 4$ | 20. $\frac{n}{7}$ |
| 4. $c - 4$ | 13. $n - 10$ | 21. $n - 3$ |
| 5. $c - 7$ | 14. $\frac{n}{6}$ | 22. n^2 |
| 6. $c - 2$ | 15. $n + 9$ | 23. $\frac{n}{9}$ |
| 7. $c - 5$ | 16. $n - 5$ | 24. $\frac{n}{10}$ |
| 8. $a - 3$ | 17. $n + 7$ | |
| 9. $a + 10$ | | |

Medium (drill)

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|--------------------|---------------------|----------------------|
| 1. $cp + 2$ | 6. $\frac{k+x}{10}$ | 11. $ab - 5$ |
| 2. $\frac{k+x}{9}$ | 7. $a^2 + 11$ | 12. $2b - 7$ |
| 3. $2k - 4$ | 8. $5a - 3$ | 13. $13y$ dollars |
| 4. $2(a + 8)$ | 9. $7a - 7$ | 14. $\frac{70}{k}$ |
| 5. $bk + 3$ | 10. $3b - 9$ | 15. $8k$ |
| | | 16. $y + 10$ minutes |

Challenging

1. $3n$ dollars
2. $2s + 4$
3. $2L + 10$

Extension

1. $3e + 6$
2. $3n + 3 = 3(n + 1)$

Exit ticket

1. Level 1: $N + 12$
2. Level 2: $y - 4$
3. Level 3: $5m + 2$

5.05 Substitution (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 0
3. Ex 2a: 60
4. Ex 2b: 109
5. Ex 3a: -11
6. Ex 3b: 94

Easy (drill)

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|-------|-------|-------|
| 1. 5 | 4. 24 | 7. 14 |
| 2. 50 | 5. 80 | 8. -2 |
| 3. 40 | 6. 45 | 9. 27 |

10. 49

11. 45

12. 16

13. 6

14. 16

15. 21

16. 3

17. 56

18. 18

19. 49

20. 11

21. 20

22. 15

23. 28

24. 13

Medium (drill)

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|-------|--------|---------------|
| 1. 12 | 9. -24 | 17. $s = 12$ |
| 2. 11 | 10. 4 | 18. $A = 121$ |
| 3. 12 | 11. 9 | 19. $C = 199$ |
| 4. 8 | 12. 2 | 20. $P = 32$ |
| 5. 9 | 13. -9 | 21. $P = 36$ |
| 6. 6 | 14. 2 | 22. $s = 10$ |
| 7. 30 | 15. 25 | |
| 8. 2 | 16. 5 | |

Challenging

1. 22
2. 35
3. 77

Extension

1. \$59
2. 60 minutes

Exit ticket

1. Level 1: 20
2. Level 2: 13
3. Level 3: 10

5.07 Adding and subtracting terms (booklet

p. 23)

WE DO examples (teacher)

1. Ex 1a: $7m$
2. Ex 1b: $8m$
3. Ex 2a: $2y$
4. Ex 2b: $18uv + 4$
5. Ex 3a: $13h^2 + 17h$
6. Ex 3b: $2h^3 - 14h$

Easy (drill)

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|----------|-----------|---------|
| 1. $5n$ | 12. $4y$ | 23. no |
| 2. $-2p$ | 13. $-2c$ | 24. no |
| 3. $4t$ | 14. $6a$ | 25. yes |
| 4. $13c$ | 15. $-5p$ | 26. yes |
| 5. $2y$ | 16. $-4x$ | 27. yes |
| 6. $12k$ | 17. yes | 28. yes |
| 7. $9m$ | 18. yes | 29. no |
| 8. $6p$ | 19. yes | 30. yes |
| 9. $21t$ | 20. yes | 31. yes |
| 10. $5x$ | 21. yes | 32. no |
| 11. $-m$ | 22. no | |

Medium (drill)

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|-------------------|-------------------|-------------------|
| 1. $-6c - 2y$ | 8. $12a - 8k$ | 14. $-mp + 4m^2$ |
| 2. $-2b - 7k$ | 9. $4c - 8y$ | 15. $-7m^2$ |
| 3. $7a - 7b + 9$ | 10. $3k - p + 9$ | 16. $4cp + c^2$ |
| 4. $6c - 9p$ | 11. $6b + x + 6$ | 17. $-5bt + 3b^2$ |
| 5. $10m + 4n + 5$ | 12. $6c - 4y$ | 18. $5bt + 6b^2$ |
| 6. $12n - 9x$ | 13. $5mp + 15m^2$ | |
| 7. $3n - 8t$ | | |

Challenging

1. $13h^2 + 17h$
2. $2h^3 - 14h$
3. $8ab + 5a - b$

Extension

1. $6a^2 + 2a - 6$
2. $6m$

Exit ticket

1. Level 1: $9k$
2. Level 2: $3a + 9$
3. Level 3: $2x^2 - 3x$

5.08 Multiplying terms (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: $12m^2$
2. Ex 1b: a^4
3. Ex 2a: $180v^2$
4. Ex 2b: $15p^2q^2$
5. Ex 3a: $21a^2$
6. Ex 3b: $\tilde{p}m^3n^3$

Easy (drill)

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|------------|------------|
| 1. $49kp$ | 13. $24k$ |
| 2. $45t$ | 14. $18p$ |
| 3. $16bc$ | 15. $56ab$ |
| 4. $24by$ | 16. $32ac$ |
| 5. $20c$ | 17. ct^2 |
| 6. $56bn$ | 18. cm^2 |
| 7. $6cp$ | 19. m^2 |
| 8. $32cp$ | 20. n^2t |
| 9. $15a$ | 21. p^2 |
| 10. $16m$ | 22. cx^2 |
| 11. $36k$ | |
| 12. $15ac$ | |

Medium (drill)

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|---------------|---------------|
| 1. $18k^2m$ | 8. $30ny$ |
| 2. $20m^2p^2$ | 9. $30my$ |
| 3. $18k^2m^2$ | 10. $6p^2y^2$ |
| 4. $20c^2$ | 11. $4nx^2$ |
| 5. $18c^2$ | 12. $10k^2$ |
| 6. $10ak$ | 13. $10bn$ |
| 7. $20a^2k^2$ | 14. $18c^2k$ |
| | 15. $12py$ |

Challenging

1. $15p^2q^2$
2. $\tilde{p}m^3n^3$
3. $6a^5$

Extension

1. $9m^2$
2. $5xy$

Exit ticket

1. Level 1: $15k$
2. Level 2: $24a^2b$
3. Level 3: $6x^2y$

5.09 Dividing terms (booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: $5m$
2. Ex 1b: 9
3. Ex 2a: $7y$
4. Ex 2b: $3x$
5. Ex 3a: $\frac{16e}{5}$
6. Ex 3b: $2xy$

Easy (drill)

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|----------|----------|---------|
| 1. $9p$ | 12. $5m$ | 23. n |
| 2. $2a$ | 13. $3p$ | 24. x |
| 3. $4t$ | 14. $6c$ | 25. c |
| 4. $6n$ | 15. $8c$ | 26. t |
| 5. $3t$ | 16. $3y$ | 27. m |
| 6. $5m$ | 17. c | 28. b |
| 7. $4k$ | 18. y | 29. t |
| 8. $6b$ | 19. y | 30. x |
| 9. $2x$ | 20. a | 31. c |
| 10. $7n$ | 21. b | 32. n |
| 11. $7x$ | 22. c | |

Medium (drill)

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|----------|-----------|----------|
| 1. $7p$ | 9. $2m$ | 17. $4p$ |
| 2. $2k$ | 10. $9k$ | 18. 3 |
| 3. $5p$ | 11. $8m$ | 19. 9 |
| 4. $3t$ | 12. $8a$ | 20. $7t$ |
| 5. $9a$ | 13. $9ab$ | 21. $7t$ |
| 6. $9ab$ | 14. $7t$ | 22. 8 |
| 7. $9t$ | 15. $3n$ | 23. $2y$ |
| 8. $5b$ | 16. $4cm$ | 24. $6n$ |

Challenging

1. $2xy$
2. 3
3. m

Extension

1. $\frac{3m}{2}$
2. $2y$

Exit ticket

1. Level 1: $3p$
2. Level 2: $4ab$
3. Level 3: $\frac{-2x2}{3}$